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For more than a decade, cloud first has been the guiding principle for companies, and the default choice for speed, flexibility and scale. Cloud computing addresses many of the limitations of on-premises hosting, particularly in terms of scalability. Unlike on-prem systems, where expanding capacity depends on time-consuming hardware purchases and lead times, the cloud enables rapid horizontal scaling on demand. Today, public cloud usage is mature, and companies have access to an unprecedented range of services, optimization capabilities, and deployment models. And the cloud landscape continues to evolve rapidly, creating new opportunities to tailor infrastructure decisions to specific business and technical needs.

However the truth has emerged that maintaining control in the public cloud has its challenges. The flexibility of the cloud comes with trade-offs: misconfigured resources or poor pricing decisions can drive costs higher than expected. Additionally, the vast range of services, tools, and configuration options plus the rapid pace of cloud changes can make cloud management complex and require constant attention and up-to-date knowledge to make the best security as well as financial decisions. Further details about the main challenges reported by companies are described in the last section.

In response to the concerns, cloud repatriation, emerged as a trend in 2018. This is the process of moving workloads and data back to on-premises and private cloud environments. It doesn't mean abandoning the cloud or reverting to slow, rigid on-premises infrastructure. Though there are a few headline examples of companies that fully repatriate to save costs and regain control, it more often means a rebalance of cloud adoption, and is part of a multi- or hybrid-cloud strategy.

With these head turning cloud trends, we saw the need to sharpen the focus on to what matters: how to be deliberate in your cloud management. Being cloud deliberate is not about being pro- or anti cloud, it's about making workload-by-workload decisions, to find the optimal fit for your business as well as for your developers. In other words, keeping the benefits of the cloud, but maximizing control.

There are a million different services and configuration options in cloud solutions, and no one-size-fits-all solution. But we wanted to provide some sensible principles and hands-on tools to help guide your choices, and hopefully help you get a sense of control in the cloud, instead of feeling like you're just following the latest hype.

Here we provide some **guiding principles** and **tools** to help you be **cloud deliberate**. We then briefly go over the main reasons why companies have problems being in control in the cloud, and wrap it up with how **you** can get started today.

How to be (cloud) deliberate?

These cloud challenges is why having a cloud deliberate approach matters. It gives you the ability to act with intention. To make this practical, we suggest a simple structure:

Principles
How to think

Toolbox
How to act

Decision framework
How to decide

Principles: How to think in a shifting landscape

The principles set direction and provide a shared architectural mindset for making decisions in a landscape where services, pricing models, and regulatory requirements change continuously. They act as reference points that reduce noise, support alignment across teams, and help ensure that cloud decisions reflect organisational priorities rather than short-term trends.

1. Apply cloud-native principles wherever workloads run: Adopt operating practices inspired by cloud-native environments: automation, infrastructure as code, self-service platforms, standardised services, and loosely coupled architectures.

Principles & Toolbox

These practices enable speed, resilience, and scalability and should guide how we design and operate systems regardless of whether workloads run in public cloud, private cloud, or on-premises environments.

2. Create clarity before action: Do all you can to ensure that teams can base decisions on an explicit understanding of your current landscape, constraints and desired outcomes, rather than assumptions, habit, or latest hype.

3. Strive for robustness and be ready to recalibrate: Design architectures that tolerate change, while favouring decisions that can be revisited when costs, regulations, or requirements change.

4. Make it easy to make the right choices: Reduce cognitive load by removing unnecessary complexity. Provide clear defaults and paved roads that guide teams toward preferred solutions, while preserving the ability to deviate when justified.

Toolbox: How to act deliberately in practice

The toolbox turns principles into everyday practice. Each element supports one or more of the guiding principles and helps teams make deliberate choices without slowing down delivery.

Tool 1: Automation over manual operations

Supports Principle 1: Be cloud-native wherever you are

Deliberate cloud management relies on repeatable and predictable operations. Manual configuration and ad hoc changes make environments harder to understand, harder to reproduce, and harder to control over time.

Automation shifts operational work from individual actions to shared definitions. Infrastructure, environments, and policies are described as code, versioned, reviewed, and deployed in a controlled way. This makes changes visible, auditable, and reversible.

By automating routine tasks such as provisioning, scaling, patching, and cleanup, organisations reduce the risk of configuration drift and human error. Automation also lowers the operational burden on teams, freeing time for work that requires judgment rather than repetition.

Automation is not about speed alone. It is a prerequisite for consistency across environments, whether workloads run in public cloud, private cloud, or on-premises setups.

Tool 2: Clarity on starting point, constraints, and desired state

Supports Principle 2: Create clarity before action

Deliberate decisions require a shared understanding of where you are starting from, what boundaries you must respect, and what you are trying to achieve. Without this clarity, teams tend to default to familiar solutions or follow momentum rather than making intentional choices.

Where you are starting from

Before selecting hosting models, services, or architectural patterns, organisations should make their current landscape visible. This includes doing a complete audit of existing systems, skills, operational practices, and technical debt that influence what is realistic and sustainable.

What boundaries you must respect

Constraints must be explicit. Regulatory requirements, data residency rules, security standards, budget limits, and contractual obligations all shape which options are viable. Making these constraints visible early reduces rework and late-stage surprises.

What you are trying to achieve

A clear desired state goes beyond technical targets. It also includes how teams are expected to work, deploy, and operate systems. When outcomes are defined up front, trade-offs become easier to reason about and decisions are easier to explain.

Toolbox continued & Decision Framework

Tool 3: Architectural patterns for hybrid and multi-cloud setups

Supports Principle 3: Strive for robustness and be ready to recalibrate

Architectural patterns for hybrid and multi-cloud setups help organisations avoid tightly coupling workloads to a single environment. The goal is not to run everything everywhere, but to ensure that critical systems are not built on assumptions that are difficult to change later.

A cloud deliberate approach assumes hybrid possibilities from the start. Different workloads can live where they fit best. Compliance-sensitive or latency-critical systems may run in private environments, while elastic or customer-facing workloads benefit from public cloud capabilities. Clear separation of concerns improves robustness and operational clarity.

Freedom of movement is a key outcome of these patterns. Designing for portability allows organisations to adapt to changes in cost, regulation, performance, or provider strategy without large, disruptive migrations. This often requires trade-offs and deliberate choices up front.

In practice, this means:

- Choosing technologies that can run across environments, whether public cloud, private cloud, or on premises
- Favouring open standards such as SQL, S3-compatible APIs, or OAuth and OIDC
- Keeping data portable by avoiding unnecessary dependence on proprietary formats or services
- Designing for reversibility by favouring replaceable components over tightly integrated solutions
- Supporting developer portability by enabling local development and testing outside the primary cloud environment

Used selectively and intentionally, hybrid and multi-cloud patterns provide robustness while preserving the ability to recalibrate as conditions change.

Tool 4: Practices for continuous optimization and review

Supports Principle 3: Strive for robustness but be ready to recalibrate

Deliberate cloud decisions are not one-time events. Costs, usage patterns, regulatory requirements, and business priorities change over time, and decisions that were appropriate at one point may become suboptimal later.

Continuous optimisation establishes a regular rhythm for reviewing workloads, architectures, and operating practices. This includes cost efficiency, performance characteristics, security posture, and compliance across all environments, not only public cloud.

Practices such as FinOps help make cost and usage visible and shared across teams. When optimisation is treated as ongoing work rather than periodic cleanup, organisations reduce waste and avoid sudden corrective actions.

Regular review also supports reversibility. By revisiting assumptions and outcomes, teams can identify when a change in placement, service choice, or operating model is warranted and act before constraints become entrenched.

Tool 5: Paved roads and clear defaults

Supports Principle 4: Make it easy to make the right choices

Paved roads mean providing your teams with supported, well-understood ways of building, deploying, and operating your systems. Paved roads represent a clearly communicated preferred path for most use cases and reflect the organisation's architectural principles, standards, and constraints.

Clear defaults reduce cognitive load. When teams know what is recommended and why, they spend less time evaluating basic options and more time solving real problems. Defaults also improve consistency across teams and systems without requiring heavy governance.

A paved road does not remove autonomy. Teams can deviate when there is a justified need, but deviations are explicit and intentional rather than accidental. This balance supports both speed and control.

In a cloud deliberate approach, paved roads make preferred decisions visible and easy to follow, helping teams act in line with strategy without constant oversight.

How to make decisions?

Decisions happen at all levels, from product teams to operations. While widespread decision-making is necessary, inconsistency and lack of visibility can cause issues. Centralizing decisions reduces autonomy, but full independence leads to fragmentation. A balanced, collective approach drawing from Andrew Harmel-Law's "Facilitating Software Architecture: Empowering Teams to Make Architectural Decisions" with added enhancements supports both speed and alignment.

Empower, Consult, Measure, Guide, Document: Our framework gives everyone authority, promotes expert collaboration, considers impact and reversibility, applies guiding principles, and records key decisions in Architectural Decision Records.

1	Empower Authority should rest as close as possible to the point of impact, with decisions made by those who have to live with the outcome. Teams need to be empowered in order to move fast, but with caution as decisions are not taken in isolation.
2	Consult Decisions have an impact. Those making a decision need to consult those that will be impacted and relevant subject matter experts. Have a process in place for getting input from affected individuals and experts, to ensure consultation before decision.
3	Measure Teams should measure their decisions using two lenses: Impact and Reversibility. • Impact: Affects multiple teams/domains or has long-lasting consequences; • Reversibility: Two-way door (easy to reverse), One-way door (hard to change). One-way decisions with a high impact require more consultation and is most likely not a decision a team can make on their own. In the other end we have two-way decisions with a local impact only. These can safely be taken by the team itself.
4	Guide Principles, standards, best practices and tech radar provide guardrails or guideposts for decisions. They set direction, reduce ambiguity, clarify what is preferred/emerging/should be avoided.
5	Document Architecture Decision Record (ADR) captures decision, context, options, trade-offs, and reasoning. ADRs create transparency, shared knowledge, help teams understand architectural landscape, and make distributed decision-making safe.

Background: Why Control Can Be Difficult in the Public Cloud

Escalating and Unpredictable Costs

The public cloud's elasticity solves a problem on-premises infrastructure cannot: when you need more capacity, you get it immediately, without hardware procurement cycles that can take weeks or months. The advantage to this built-in elasticity is huge, and the pay-per-use pricing model can be cost-effective for elastic workloads. However, as operations scale, it also carries with it a risk of escalating costs that are easy to miss and hard to forecast.

Wrongly configured auto-scaling can cause data egress and network fees to accumulate quietly, or leave idle resources running over time, causing an unnecessary cost. Other sources of unnecessary costs are suboptimal pricing choices that don't leverage reserved instances or spot pricing. These costs can largely be avoided by correctly configuring and managing your system, but it requires rigorous monitoring and tracking processes, and a high-level system knowledge.

Organizations must also consider the long-term implications of workload and data placement. While applications and containers are relatively easy to move between environments, data is significantly harder and more expensive to transfer. Data egress charges can become a major barrier to repatriating workloads back on-premises or moving them between cloud providers. As datasets grow, these costs can create a form of operational lock-in that limits flexibility and complicates infrastructure strategy decisions.

Even though organizations adopt FinOps practices and dedicated teams to optimize cloud spending, cost control and predictability consistently rank as the top cloud management challenge.

Security Challenges

Public cloud providers invest heavily in security, offering advanced features such as DDoS protection, Identity and Access Management (IAM) and encryption as a standard. These are far more granular and automated security controls, which would be extremely expensive and complex to build yourself.

Cloud platforms come with a shared responsibility model; providers secure the underlying infrastructure, while customers are responsible for data, IAM and configuration. Though the security controls as mentioned are highly advanced and robust, the shared responsibility model also introduces risk. Security misconfigurations and misunderstood responsibility boundaries can lead to security gaps, such as unintended data exposure. Sustaining an up-to-date knowledge of the responsibility boundaries and configuration options in the cloud is critical to maintaining a strong security posture.

Compliance & Data Sovereignty

The public cloud can both strengthen and complicate compliance. On the positive side, major providers design their platforms to support regulatory requirements such as GDPR and HIPAA. This provides pre-audited infrastructure, built-in security controls, and tools such as encryption, logging, and region selection, which help organizations both meet and demonstrate compliance more easily.

However, due to the shared responsibility model and constant platform evolution also leads to regulatory risk. Careful oversight is required to avoid misconfigurations and misunderstandings which can lead to compliance gaps, such as overly permissive IAM policies, data exposure or data residency violations.

Performance & Latency

Public cloud platforms deliver powerful, globally distributed infrastructure and major advantages like rapid scaling and low-latency access. However, performance can vary due to multi-tenancy and shared resources, where spikes from other tenants can monopolize shared capacity, aka the "noisy neighbour" effect, causing higher latency or lower throughput. Performance issues can also arise from suboptimal configuration, such as poorly distributed workloads across regions or misconfigured auto-scaling, both of which can degrade overall performance.

That said, hybrid cloud environments can introduce additional latency when workloads and data are split between cloud and on-premise systems. Communication relies on network connectivity, which can add delays for real-time or latency-sensitive applications, and "data gravity" can further complicate things by making large datasets difficult to move, effectively pulling applications closer to where the data is stored. But hybrid setups can also improve performance by keeping critical workloads on-premise while using the cloud for scalability and less time-sensitive tasks.

Ultimately, both cloud and hybrid approaches have trade-offs, and the key is choosing the right setup based on workload and performance needs.

Vendor Lock-in

Vendor lock-in occurs because cloud platforms use proprietary tools, storage formats, or tightly integrated services such as managed databases, serverless functions, PaaS apps or Managed Machine Learning offerings. This creates dependencies to the provider's ecosystem, and makes switching difficult and costly. Organizations can become strategically bound by the provider's priorities and pace of innovation, and make them vulnerable to pricing changes, service outages or policy updates.

Strategic and Geopolitical Concerns

More organizations seek to transition away from U.S.-based cloud providers due to data sovereignty concerns and exposure to US legal, political, and technical risks. Progress is slow, however, because hyperscalers like Microsoft, AWS and Google dominate the market, while European providers lag in scale, maturity and investment. Growing concern over reliance on a few tech giants has prompted national and corporate strategies focusing on digital independence.

Cognitive Load

It's important to recognize the cognitive cost associated with the Public Cloud management challenges described above. The effort required for Cloud Management teams to keep pace with rapidly evolving platforms, and ensure that they make the right architectural and financial decisions when dealing with hundreds of services and tools, complex configurations, and continuously changing security can be significant.

At the same time, hybrid strategies introduce operational complexity of their own. Organizations running both cloud and on-premise infrastructure must maintain expertise across two distinct operational models, combining cloud-native platforms with traditional infrastructure management. This can make it more difficult to recruit and retain talent, and the need to maintain split competencies across teams can also increase the risk of operational errors, as fewer individuals fully understand the entire environment end-to-end.

The question is therefore not whether complexity exists, but where it resides and whether the organization is equipped to manage it effectively, both technically and organizationally.



Your turn

Rather than chasing every cloud trend, focus on small, intentional actions to make progress. Start by applying one or two guiding principles to real decisions for example, make optimization ongoing or favor automation over manual tasks. Build from there: set clear standards, use tools like Tech Radar to track your current state, and document key decisions with Architecture Decision Records for better alignment.

If you want support, we offer complimentary options: a casual lunch meeting to discuss your challenges or a half-day workshop to identify opportunities and outline next steps helping you move intentionally toward a lasting cloud strategy.